

Group Project: Manual Replication of Frye and Borisova (2019)

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1. Introduction

1.1 Defining the Parameters of the Replication

The report details the replication of Frye and Borisova’s (2019) article, “Elections, Protest, and Trust in Government: A Natural Experiment from Russia,” published in the *Journal of Politics*. The primary objective was to manually replicate the authors’ core models to assess their claims regarding the impact of the December 2011 post-parliamentary elections protests on political trust among residents of Moscow.

The replication is split into two main phases. The first phase focuses on re-running the original analysis, including all primary tables, conducting exploratory and confirmatory factor analysis on the government trust index, and testing alternative measures of the dependent variable. The second phase subjects the paper’s model to placebo timing tests, alternative control specifications, and worst-case non-response models to evaluate the validity of the paper’s underlying assumptions.

1.2 Declaration of AI Use

As this is primarily a manual replication, generative AI such as Claude was used for two specific purposes:

1. To compile a survey codebook using partial variable labels, as the replication material lacked a substantive English version of the codebook.
2. Since the original replication material was written in Stata, AI was used to explain, in plain language, what a line(s) of Stata code was attempting to execute. This was only used at a high level for Stata code that could not be easily discerned by the research team, and was only used to inform how the replication should be approached in R. At no point in the replication process did the manual team use AI to generate any R code.

2. Translation Process

The original replication files were coded using Stata, but required translation into R, which is the primary programming and statistical language used by the research team.

2.1 Building the Codebook

A significant early challenge was that no survey codebook was included in replication materials. Dataset variables were simply named by their question ID numbers, with labels consisting of the original questions in Russian. Since those labels were truncated at 80 characters, the research team obtained incomplete questions. This was especially problematic for questions that repeated the same setup for different sub-items (e.g. question q9a-q9j). To overcome this challenge, the research team used Claude AI to infer the missing information and reconstruct a functional codebook.

2.2 Translating from Stata to R

One of the first challenges the research team encountered was translating the Stata code into R, as resource limitations meant the research team had no access to Stata. An option that was explored early on was using the **RStata** package to execute Stata files directly into **RStudio**, but this required installing the Stata language, which the research team was unable to do.

To overcome this challenge, the research team relied on a combination of the Claude-built codebook, the Appendix documents in the supplementary materials available on the *University of Chicago Press Journals*, and Claude outputs that explained high-level Stata code to the research team. For example, Figure 2.1 illustrates a case where the research team was confident in manually translating the Stata code into R, whereas Figure 2.2 illustrates a case where the research team relied on the aforementioned methods for translation.

```
*trust

gen trustarmy = q9a
recode trustarmy 99 =. 1=5 2=4 4=2 5=1

gen trustpolice = q9b
recode trustpolice 99=. 1=5 2=4 4=2 5=1

gen trustfsb = q9c
recode trustfsb 99=. 1=5 2=4 4=2 5=1

gen trustcourt = q9d
recode trustcourt 99=. 1=5 2=4 4=2 5=1

gen trustmungov = q9e
recode trustmungov 99=. 1=5 2=4 4=2 5=1
```

Figure 2.1: Stata code sample: Variable recoding

Stata has certain functionalities that it automatically executes, which the research team had to overcome when translating the code into R and running the replication analysis. First,

Stata automatically omits collinear predictor variables from its analysis. This specifically affected the `okrug` 10 variable ($n = 29$), which had no United Russia respondents in the post-election window. In particular, this made the United Russia $\times$ Post-Election interaction a constant that had to be manually fixed. Consequently, Stata’s silent omission meant that the published results gave no indicator that a predictor had been dropped. Conversely, R’s explicit error forced the research team to diagnose and document the issue.

```
*Table 2
reg trustgovind period3 period2min protestday electday wealth education okrugdv1-okrugdv10, robust
outreg2 using tab2, keep(period3 period2min protestday electday) dec(2) alpha(0.001, 0.05) symbol(*,*) label re
reg trustarmy period3 period2min protestday electday wealth education okrugdv1-okrugdv10, robust
outreg2 using tab2, keep(period3 period2min protestday electday) dec(2)alpha(0.001, 0.05) symbol(*,*) label
reg trustpolice period3 period2min protestday electday wealth education okrugdv1-okrugdv10, robust
```

Figure 2.2: Stata code sample: Regressions and table outputs

Second, there were different code formats for some variable transformations. This required, for example, creating individual dummy variables for each level of the `okrug` variable. It should be noted that the inclusion of dummy variables may have downstream effects on the regression results, particularly the reported R^2 values. However, the coefficients on the substantive variables of interest were unaffected in direction and statistical significance.

Third, missing values (NAs) had to be handled more carefully and explicitly in R, in particular, requiring separate conditions to ignore missing values within functions like `ifelse()`. This was important for recovering the sample sizes reported in the paper’s replication material appendices.

2.3 Inspecting the Data

A review of the dataset in the replication materials indicates that it is consistent with the dataset used in the original paper. In particular, the total observations ($n = 1,550$) in the dataset matched what was in the article. Likewise, the number of per-day interviews was consistent with those listed in the Supplemental Appendix 1, which was recovered after tabulating the interview data variable against Appendix 1.

The total number of variables was 292. There did not appear to be any duplicate IDs. There were no unrealistic values in the age range, with all respondents being aged 18 to 85.

The results in the Appendix generally matched what was in the paper, with some caveats. For the tables listed under Appendix 2A, the number of observations is listed as 1,232. However, the replication material had 1,521 observations, a mismatch likely due to Stata automatically dropping collinear items. Similarly, the tables listed under Appendix 2B were virtually the same, with the number of observations differing from 1,232 in the supplementary materials and 1,521 in the dataset. Appendices 3.1 and 3.2 had the same results as the paper, with the number of observations matching exactly.

Much of the concern regarding data integrity lies with how the data was actually collected, as the dataset available for replication had several quality issues. This included incompleteness, difficulty identifying what some objects and variables refer to, and substantive amounts of data being collected differently across the sample. For instance, certain partisanship battery questions were only administered to pre-election respondents, limiting their use in post-protest comparisons.

There were two key issues the research team identified when inspecting the data. First, the Federal Security Service (FSB) non-response rate triples from 4% to 15% post-protest. The concern for the research team here is that post-protest FSB trust estimates may reflect a compositional change among respondents, with fearful respondents opting out, rather than genuine attitudinal change. Similarly, the United Nations non-response doubles from 11% to 24%. Second, there are 318 (21%) missing observations from the initial trust index regression.

3. Re-Running the Original Analysis

3.1 Replicating the Core Analysis

The research team was able to successfully replicate the Determinants of Trust Institutions, such that the coefficients matched the statistical significance and direction of the results in the original paper.

	Gov Index	Army	Police	FSB	Court	Municipal Gov	Federal Gov	Duma	Procuracy	United Nations
Post-protest	0.17** (0.05)	0.19** (0.07)	0.24*** (0.06)	0.23*** (0.07)	0.13* (0.06)	0.19** (0.06)	0.18** (0.07)	0.08 (0.07)	0.05 (0.06)	0.06 (0.07)
Post-election	0.06 (0.06)	0.25*** (0.07)	0.20** (0.07)	0.13+ (0.07)	0.11 (0.07)	-0.01 (0.07)	-0.04 (0.07)	-0.01 (0.07)	-0.03 (0.07)	-0.11 (0.07)
Protest day	0.14 (0.10)	0.49*** (0.11)	0.36*** (0.10)	0.25* (0.12)	0.22* (0.11)	0.15 (0.12)	0.07 (0.12)	-0.06 (0.12)	-0.06 (0.12)	0.05 (0.12)
Election day	-0.02 (0.12)	-0.10 (0.16)	0.20 (0.16)	0.18 (0.17)	0.12 (0.15)	-0.19 (0.14)	-0.28+ (0.15)	-0.18 (0.14)	0.01 (0.15)	-0.49** (0.17)
Num.Obs.	1232	1491	1505	1338	1450	1485	1487	1466	1437	1212
R2	0.088	0.072	0.057	0.087	0.057	0.072	0.085	0.062	0.053	0.042

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Figure 3.1: Replicated table: Determinants of trust in institutions

3.2 Factor Analysis

Exploratory (EFA) and confirmatory (CFA) factor analysis were conducted on all eight trust variables, with similar factor loading scores using the `psych` and `lavaan` packages in R. The results yielded a Cronbach's alpha of ≈ 0.903 , both raw and standardized, indicating high internal consistency and high reliability. The underlying implication is that the government trust index in the appears to capture the same underlying latent variable: trust in government.

variable	efa.loading.score	cfa.loading.score
trustarmy	0.5296033	0.5288245
trustpolice	0.6906914	0.6932360
trustfsb	0.7031063	0.6874910
trustcourt	0.6796468	0.6942999
trustmungov	0.8200351	0.8422222
trustfedgov	0.8462218	0.8723872
trustduma	0.8168069	0.8414339
trustprocuracy	0.7175109	0.7246028

Figure 3.2: Replicated table: Determinants of trust in institutions

3.3 Alternative Trust Measures and Indices

In addition to re-running the main analysis and conducting factor analysis, the research team investigated alternative ways of measuring the dependent variables. The first involved including the Orthodox church in the trust index, which was the only institution from the battery of trust items that was omitted from the index. Doing so yielded a statistically significant coefficient on the post-protest variable at $p < 0.05$.

	Gov Index (with Church)	Church
Post-protest	0.17** (0.05)	0.13* (0.06)
Post-election	0.06 (0.06)	0.05 (0.07)
Protest day	0.12 (0.09)	0.00 (0.12)
Election day	-0.04 (0.12)	-0.27 (0.20)
Num.Obs.	1192	1445
R2	0.087	0.042

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Figure 3.3: Accounting for the role of the Church

The second involved dropping one factor at a time from the trust index. The results similarly yield a statistically significant post-protest coefficient at $p < 0.05$.

	No Army	No Police	No FSB	No Court	No Municipal Gov	No Federal Gov	No Duma	No Procuracy
Post-protest	0.14** (0.05)	0.13** (0.05)	0.11* (0.05)	0.13** (0.05)	0.12* (0.05)	0.12* (0.05)	0.12* (0.05)	0.14** (0.05)
Post-election	0.08 (0.10)	0.09 (0.09)	0.12 (0.08)	0.10 (0.09)	0.13 (0.09)	0.14 (0.09)	0.13 (0.09)	0.08 (0.09)
Protest day	0.16 (0.14)	0.18 (0.13)	0.22+ (0.12)	0.20 (0.13)	0.24+ (0.13)	0.26* (0.13)	0.27* (0.13)	0.21 (0.13)
Election day	-0.03 (0.12)	-0.07 (0.12)	-0.09 (0.12)	-0.07 (0.12)	-0.03 (0.12)	-0.03 (0.12)	-0.03 (0.12)	-0.05 (0.12)
Num.Obs.	1242	1235	1332	1243	1235	1236	1242	1260
R2	0.082	0.088	0.078	0.089	0.090	0.089	0.088	0.087

Figure 3.4: Regression models using leave-one-out indices

The third was sub-indices based on alternative concepts of trust, in particular, those

derived from Schneider (2017). Again, with the exception of Model 2.1, the results yield statistically significant post-protest coefficients at $p < 0.05$.

	Model 1	Model 2.1	Model 2.2	Model 3.1	Model 3.2
Post-protest	0.16* (0.06)	0.18*** (0.05)	0.19*** (0.05)	0.17** (0.05)	0.22** (0.07)
Post-election	-0.02 (0.06)	0.11* (0.06)	0.09 (0.06)	0.06 (0.06)	0.08 (0.07)
Protest day	0.07 (0.11)	0.22* (0.09)	0.20* (0.09)	0.12 (0.10)	0.16 (0.12)
Election day	-0.22+ (0.13)	-0.08 (0.13)	-0.01 (0.13)	-0.03 (0.12)	0.03 (0.15)
Num.Obs.	1438	1417	1282	1388	1275
R2	0.079	0.086	0.095	0.070	0.082
Fed Gov	Yes	Yes	Yes	Yes	Yes
Duma	Yes	Yes	Yes	Yes	Yes
Mun Gov	Yes				
Police		Yes	Yes	Yes	Yes
Army		Yes	Yes		
Court				Yes	Yes
FSB			Yes		Yes

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Figure 3.5: Sub-indices constructed following Schneider (2017)

The fourth alternative measure was the loading score-weighted trust index, where the research team again found a statistically significant post-protest coefficient at $p < 0.05$.

	(1)
Post-protest	0.16** (0.05)
Post-election	0.05 (0.06)
Protest day	0.12 (0.10)
Election day	-0.03 (0.12)
Num.Obs.	1232
R2	0.085

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Figure 3.6: Factor loading score-weighted index

4. Further Tests and Robustness Checks

4.1 Placebo Tests

The paper identifies December 10th, the date of the largest protest, as a sharp treatment date. However, a visualization of the daily mean trust in government suggests this assumption may be problematic. Rather than a discontinuous break in daily means on December 10th, the data reveals a noisy but gradual upward trend in the weeks before the protest:

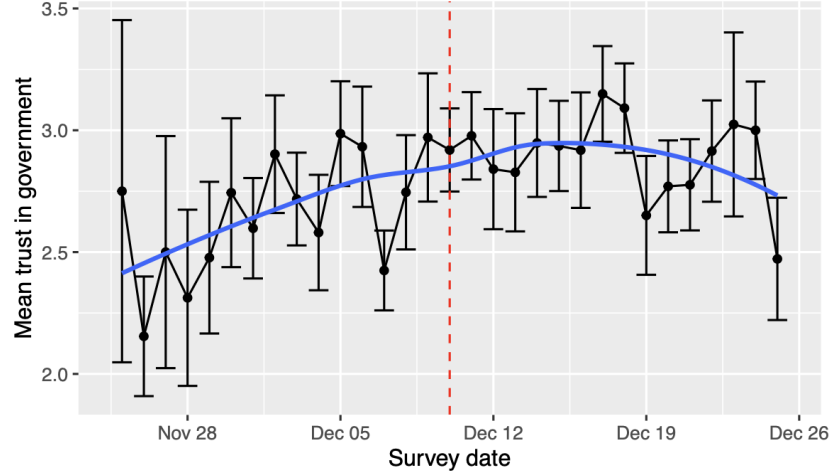


Figure 4.1: Daily mean trust in government, with 95% CIs and Loess smoothing

To test whether the December 10th protest truly caused a shift in attitudes toward the government, the research team conducted a series of placebo timing tests. If this assumption of the protest having a sharp, causal impact is correct, then artificially shifting the treatment date to a time before or after the protest should yield a weak or null effect. Alternatively, detecting a significant effect before or after the protest occurred would indicate that the original paper captures an underlying time trend rather than the isolated shock caused by the protest itself. Table 4.2 applies this hypothesis:

Placebo Timing Tests		
Post-treatment coefficients using alternative protest dates		
Date	Post Coef.	p-value
Dec 06	0.067	0.202
Dec 07	0.077	0.142
Dec 08	0.148	0.003
Dec 09	0.146	0.002
Dec 10	0.140	0.003
Dec 11	0.117	0.013

Figure 4.2: Post-treatment coefficients for *Trust in Government Index* using alternative protest dates: December 8th, 9th, and 10th

As Table 4.2 demonstrates, artificially setting the treatment date to December 8th or 9th yields post-treatment coefficients that are nearly identical to the real protest date, December 10th. Because the treatment effect appears before the treatment itself, the paper's assumptions of no pre-treatment effects and correct treatment timing are violated. Rather than identifying a discontinuous shock caused by the December 10th protest, the paper's model appears to pick up the broader pattern of shift that began at least two weeks earlier, in late November.

4.2 Manipulating Controls

The research team re-estimated the model with varying control specifications with no controls, baseline controls, and additional covariates to check whether day-to-day differences in the sample affected outcomes. Ultimately, the research team obtained results largely similar to the original. Running the model without any controls yielded a larger, statistically significant post-protest coefficient of 0.254. Running the paper’s baseline model yielded a highly statistically significant result of 0.168. Finally, running the main regression with all the dataset’s controls (age, gender, permanent residency status, nationality, unemployment status, and state-sector employment) yielded a statistically insignificant result of 0.085. While the results are largely robust to the removal of controls, the loss of significance under the full-control regime shows sensitivity to model specification and possible confounding.

4.3 Non-Response Analysis and Worst-Case Modelling

As noted, one of the concerns that emerged from investigating the data integrity was the fact that the FSB non-response rate jumped from 4% (pre-election) to 15% (post-protest). Similarly, the UN non-response rate jumps from 11% (pre-election) to 24% (post-protest). The concern is that if respondents who declined to rate the FSB and UN post-protest are disproportionately distrustful, then the observed post-protest trust increase is upwardly biased. Consequently, as a robustness measure, the research team reran Appendix Table 2A under the assumption that all non-respondents are maximally distrustful.

	Gov't Index	Army	Police	FSB	Court	City Gov't	Fed Gov't	Duma	Procu racy	UN
	1	2	3	4	5	6	7	8	9	10
Post-Protest	.16* (.06)	.23* (.08)	.28* (.07)	.27* (.08)	.13 (.07)	.17* (.07)	.17* (.08)	.05 (.08)	-.01 (.08)	.05 (.09)
Post-Election	.04 (.07)	.28* (.09)	.24* (.08)	.18* (.09)	.08 (.09)	-.04 (.08)	-.05 (.09)	-.06 (.09)	-.14 (.09)	-.11 (.10)
Protest day	.13 (.10)	.50* (.12)	.38* (.11)	.28* (.12)	.20 (.11)	.14 (.12)	.07 (.12)	-.09 (.12)	-.12 (.12)	.05 (.13)
Election day	-.03 (0.12)	-.09 (0.16)	0.22 (0.16)	0.21 (0.17)	0.11 (0.15)	-.021 (0.15)	-.028 (0.15)	-.021 (0.14)	-.04 (0.15)	-.049* (0.17)
Constant	3.18 (.17)	3.94 (.20)	2.99 (.18)	2.30 (.26)	2.95 (.18)	3.35 (.19)	3.50 (.20)	3.06 (.20)	3.05 (.25)	2.91 (.24)
F-test: Post-Protest = Post-Election	3.93	.65	.41	1.49	.45	10.1	11	2.95	3.46	3.93
Prob > F	.05	.4	.5	.2	.5	.001	.001	.09	.06	.05
Obs	1232	1491	1505	1338	1450	1485	1487	1466	1437	1212
R-squared	.09	.07	.06	.09	.06	.07	.09	.06	.05	.04

Figure 4.3: Regressions using worst-case modelling

Rerunning the analysis as described above causes the FSB coefficient to fall from 0.273 to 0.076 and to lose its statistical significance. Similarly, the analysis causes the UN coefficient to drop from 0.5 to -0.159, with neither the original nor the replication results being statistically significant. Conversely, the Army, Police, and City Government trust results remained robust under the worst-case scenario imputation. In sum, the FSB loses its statistical significance under the worst-case scenario.

4.4 Partial Response Trust Index

In the original paper, 318 observations constituting 21% of the sample are dropped from the trust index regression. The primary concern is that this attrition is not random, and thus, the research team check the robustness of the results by investigating whether this loss is driven by missing trust items or missing control variables.

To test this, the research team recomputed the government trust index as the mean of whichever items each respondent answered (rather than requiring all eight survey answers). Re-running the model this way revealed zero additional observations relative to the original model. This partial response robustness check therefore confirms that the 21% sample attrition in the trust index model is driven entirely by missing control variables rather than missing trust items.

4.5 Heterogenous Effects by Education and by State Sector Employment

To further investigate the mechanics behind the post-protest shifts in attitudes, the research team expanded the original paper’s analysis to test for heterogeneous effects across different subgroups, specifically focusing on education levels and state sector employment.

First, to determine if the post-protest increase in trust was concentrated among specific educational brackets, we recoded the continuous education variable into a categorical variable representing four tiers of educational attainment and interacted these tiers with the event periods. The results (in Table 4.4) indicate that while baseline trust in government is significantly lower among the highest-educated respondents (Education Group 3), the post-protest increase in trust was broadly consistent across all education tiers. Specifically, the interaction terms between the educational brackets and the post-protest indicator (“Educ Group 2 x Post-Protest” and “Educ Group 3 x Post-Protest”) yielded no statistically significant coefficients across the government trust index or any individual state institution. This suggests that a respondent’s educational background did not meaningfully alter their reaction to the December 10th events.

Second, we assessed whether a respondent’s employment in the state sector altered their reaction to the protests. To this end, we interacted a state-sector employment dummy variable with the post-election and post-protest indicators. The results, outlined in Table 4.5, reveal a significant divergence in trust. The baseline Post-protest coefficient (representing respondents who do not work in the state sector) remains positive and statistically significant across several institutions, including the police (0.20), FSB (0.20), Municipal Government (0.16), and Federal Government (0.16).

	Gov Index	Army	Police	FSB	Court	Municipal Gov	Federal Gov	Duma	Procuracy	United Nations
Post-protest	0.16 (0.17)	0.26 (0.18)	0.37* (0.16)	0.24 (0.19)	0.09 (0.16)	0.24 (0.18)	0.16 (0.18)	0.10 (0.18)	0.14 (0.17)	0.08 (0.22)
Post-election	-0.01 (0.20)	0.27 (0.19)	0.15 (0.18)	0.21 (0.22)	-0.02 (0.19)	-0.20 (0.18)	0.02 (0.20)	-0.07 (0.19)	-0.12 (0.20)	-0.48+ (0.27)
Protest day	0.13 (0.10)	0.49*** (0.11)	0.34** (0.10)	0.23+ (0.12)	0.20+ (0.11)	0.14 (0.12)	0.07 (0.12)	-0.06 (0.12)	-0.06 (0.12)	0.05 (0.12)
Election day	-0.02 (0.12)	-0.09 (0.16)	0.20 (0.15)	0.18 (0.17)	0.12 (0.15)	-0.19 (0.14)	-0.27+ (0.15)	-0.18 (0.14)	0.02 (0.15)	-0.48** (0.17)
Education group 2 (4-6)	-0.16 (0.15)	0.01 (0.17)	0.02 (0.15)	-0.08 (0.17)	-0.12 (0.15)	-0.25 (0.17)	-0.34* (0.16)	-0.33+ (0.17)	-0.10 (0.16)	-0.35+ (0.20)
Education group 3 (7-9)	-0.36* (0.16)	-0.11 (0.18)	-0.21 (0.15)	-0.34+ (0.18)	-0.35* (0.16)	-0.48** (0.17)	-0.44** (0.17)	-0.43* (0.17)	-0.24 (0.16)	-0.25 (0.20)
Educ Group 2 x Post-Protest	-0.09 (0.18)	-0.11 (0.19)	-0.24 (0.17)	-0.15 (0.20)	-0.06 (0.17)	-0.22 (0.19)	-0.06 (0.19)	-0.06 (0.19)	-0.15 (0.18)	0.01 (0.23)
Educ Group 3 x Post-Protest	0.15 (0.18)	-0.03 (0.20)	-0.02 (0.18)	0.19 (0.21)	0.19 (0.18)	0.16 (0.19)	0.17 (0.19)	0.04 (0.20)	-0.03 (0.18)	-0.07 (0.24)
Educ Group 2 x Post-Election	0.10 (0.21)	0.05 (0.20)	0.08 (0.19)	-0.08 (0.24)	0.15 (0.21)	0.14 (0.20)	-0.01 (0.22)	0.12 (0.20)	0.12 (0.22)	0.57* (0.28)
Educ Group 3 x Post-Election	0.02 (0.22)	-0.15 (0.21)	0.01 (0.20)	-0.13 (0.25)	0.10 (0.21)	0.28 (0.21)	-0.16 (0.23)	-0.02 (0.21)	0.06 (0.23)	0.18 (0.29)
Num.Obs.	1232	1491	1505	1338	1450	1485	1487	1466	1437	1212
R2	0.097	0.072	0.059	0.094	0.062	0.089	0.097	0.072	0.056	0.055

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Figure 4.4: Heterogeneous effects on trust by education group

	Gov Index	Army	Police	FSB	Court	Municipal Gov	Federal Gov	Duma	Procuracy	United Nations
Post-protest	0.11+ (0.06)	0.11 (0.08)	0.20** (0.07)	0.20** (0.08)	0.07 (0.08)	0.16* (0.07)	0.16* (0.08)	0.01 (0.08)	0.00 (0.08)	0.04 (0.09)
Post-election	0.03 (0.07)	0.19* (0.08)	0.19* (0.08)	0.08 (0.08)	0.05 (0.08)	-0.02 (0.08)	-0.08 (0.08)	-0.09 (0.08)	-0.03 (0.08)	-0.14 (0.09)
Protest day	0.01 (0.10)	0.45*** (0.13)	0.29* (0.12)	0.14 (0.14)	0.07 (0.12)	-0.02 (0.14)	-0.08 (0.13)	-0.27* (0.13)	-0.20 (0.13)	-0.05 (0.14)
Election day	0.02 (0.14)	-0.14 (0.17)	0.32+ (0.18)	0.35+ (0.19)	0.05 (0.17)	-0.17 (0.16)	-0.20 (0.17)	-0.21 (0.16)	0.08 (0.17)	-0.46* (0.19)
State sector	0.42 (0.33)	-0.18 (0.33)	0.17 (0.34)	0.42 (0.46)	0.09 (0.30)	0.64 (0.41)	0.46 (0.32)	0.33 (0.36)	0.08 (0.26)	0.48 (0.36)
State sector x Post-Protest	-0.74* (0.37)	0.01 (0.41)	-0.32 (0.42)	-0.53 (0.50)	-0.35 (0.37)	-0.89* (0.45)	-0.67+ (0.36)	-0.90* (0.39)	-0.58+ (0.31)	-1.39*** (0.41)
State sector x Post-Election	-0.25 (0.36)	0.28 (0.39)	0.00 (0.41)	-0.36 (0.49)	-0.07 (0.37)	-0.24 (0.44)	-0.21 (0.37)	-0.22 (0.39)	-0.21 (0.31)	-0.06 (0.39)
Num.Obs.	867	1023	1037	937	1005	1019	1020	1009	997	851
R2	0.116	0.080	0.063	0.113	0.059	0.086	0.097	0.084	0.075	0.051

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Figure 4.5: Heterogeneous effects on trust by state sector employment

5. Conclusion

The research team conducted a manual replication of the original paper’s core analysis and appendices, translating the project entirely from State to R in the process. By recreating the Stata model, the team confirmed that the re-estimated table coefficients matched in magnitude, direction, and significance with those from the paper. The original paper’s baseline findings held up well under initial stress tests. Exploratory and confirmatory factor analysis verified that all eight variables in the trust index accurately capture a single latent variable of trust in government. Alternative measures of the dependent variable yielded results consistent with the original paper’s claims. However, the process of translating the code and running deeper robustness checks ultimately revealed significant problems in the paper’s data integrity and identification strategy.

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